

Oversampling and Undersampling

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0.1 Introduction

Machine learning models perform best when the dataset is balanced—meaning each class in the target variable has roughly the same number of samples.

However, in real-world scenarios like fraud detection, disease classification, churn prediction, or anomaly detection, one class often appears far less frequently than the other. This situation is known as **class imbalance**, and it can cause models to become biased toward the majority class.

This imbalance leads to:

- **Poor Model Generalization:** The model is biased toward the majority class.
- **Skewed Evaluation Metrics:** Accuracy becomes misleading as the model often predicts the majority class.
- **Limited Learning of Minority Class Patterns:** The minority class may not contribute enough to the training process.

To address this, we use two powerful techniques: oversampling and undersampling.

0.2 Oversampling for machine learning

Now we will talk more specifically about oversampling data for machine learning models. We will start by discussing what oversampling is and how oversampling is performed. After that, we will talk about the advantages and disadvantages of oversampling for machine learning.

0.2.1 What is oversampling?

Oversampling is a resampling scheme where you modify the distribution of a variable in your dataset by artificially increasing the number of observations that take on a particular value or range of values for that variable. In most cases, this is done by looking at what values are underrepresented in the dataset and artificially increasing the number of observations that take on that value or range of values.

There are two different types of methods that can be used to perform oversampling. The first type of method works by duplicating existing entries that are already present in the dataset to increase the presence of those entries. The second type of method works by adding some noise to the entries that already exist and creating new “synthetic” observations that resemble the existing observations.

0.2.2 Advantages of oversampling for machine learning

Let's go through the main advantages that oversampling has over other resampling schemes.

- **Does not decrease the size of the dataset.** One advantage that oversampling has over other resampling schemes is that it does not decrease the size of the dataset. This

can be important if you are using a complicated model with many parameters that need to be fit as it ensures that your model will have sufficient data to train on.

- **Does not lose any information.** Another advantage of oversampling is that it ensures that you do not lose any information that is already present in the dataset. This is because all of the records that are contained in the initial dataset are also contained in the resampled dataset.

0.2.3 Disadvantages of oversampling for machine learning

Here are some of the main disadvantages of oversampling for machine learning.

Can increase the chance of overfitting. One disadvantage of oversampling for machine learning is that it can increase the chance of your model overfitting to your training data. Whether you perform oversampling by introducing duplicated observations or synthetic observations that closely resemble existing observations, you are increasing the presence of characteristics in your dataset that were originally only applicable to a small number of observations. These characteristics might be specific details of a few specific observations rather than broadly applicable trends.

0.3 Undersampling for Machine Learning

Now we will talk about undersampling data for machine learning models. We will start out by describing what undersampling is and how undersampling works. After that, we will discuss some of the main advantages of undersampling. Finally, we will discuss some of the main disadvantages of undersampling.

0.3.1 What is undersampling?

Undersampling is a resampling scheme where you modify the distribution of a variable in your dataset by artificially decreasing the number of observations that take on a particular value or range of values for that variable. This is done by looking at what values are overrepresented in the dataset and decreasing the number of observations that take on that value or range of values.

0.3.2 Advantages of undersampling for machine learning

Here are some of the main advantages that undersampling has compared to oversampling.

No need to introduce redundant information into your dataset. The main advantage that undersampling has is that you do not have to add any artificial observations to your dataset that introduce repeated or redundant information into your dataset. Why is this beneficial? This is beneficial because when you duplicate existing observations (or create close analogs of existing observations), you are making it seem like the patterns that are seen in those observations are more widespread than they really are. This can lead to things like models overfitting to specific patterns that were only seen in a few observations in the original dataset.

0.3.3 Disadvantages of undersampling for machine learning

What are some of the main disadvantages of undersampling for machine learning? Here are some of the main disadvantages that undersampling has compared to oversampling.

- **Reduces the size of your dataset.** The first disadvantage of undersampling for machine learning is that it reduces the size of your dataset. Machine learning models

generally perform better when they are trained on larger datasets with more observations, so this can have negative effects on the predictive performance of your model.

- **Loses information.** The next disadvantage of undersampling is that there is some loss of information. When you permanently remove observations from your dataset, you will naturally lose the information that was contained within those observations.

0.4 Conclusion

Handling imbalanced data is crucial for building reliable and fair machine learning models. Both **undersampling** and **oversampling** are effective, but each has strengths and limitations. Choosing the right technique depends on dataset size, model complexity, and the importance of preserving information.

Balancing your dataset correctly can significantly improve model accuracy, precision, recall, and overall performance—especially for critical tasks like fraud detection or healthcare analytics.

0.5 Resources:

1. [https://crunchingthedata.com/oversampling-vs-undersampling/Oversampling vs undersampling for machine learning](https://crunchingthedata.com/oversampling-vs-undersampling/Oversampling-vs-undersampling-for-machine-learning) - Crunching the Data
2. <https://www.datasciencecentral.com/exploring-oversampling-and-under-sampling-core-techniques-for-balancing-imbalanced-datasets-in-ml/> Exploring oversampling and under-sampling: Core techniques for balancing imbalanced datasets in ML - DataScienceCentral.com